This tool classifies GitHub bug reports as performance bug related or not using two approaches: a Naive Bayes + TF-IDF baseline and a fine tuned DistilBERT model. Results including metrics, statistical tests, box plots and cross project generalisation are saved automatically.

Step 1 — Clone the Repository

git clone https://github.com/msm-5x9/ISE-CW.git

cd ISE-CW

Step 2 — Install Dependencies

pip install torch transformers scikit-learn pandas numpy matplotlib scipy nltk

Step 3 — Download the Datasets Download the five CSV files from the lab GitHub repository:

https://github.com/ideas-labo/ISE/tree/main/lab1

Place all five CSV files in the same directory as br_classification_full.py (already provided).

Step 4 — Run the Tool

python3 br_classification_full.py

Step 5 — View Results All outputs are saved to the results/ directory automatically created in the working directory:

Output	Description
results_summary.csv	Median, IQR, p-value, A12 per project per metric
results/<project>_raw_scores.csv	All 30 raw scores per run per model
results/cross_project_results.csv	Cross-project generalisation results
results/plots/<project>_boxplots.png	Box plots per project
results/plots/<project>_histograms.png	Histograms per project
results/plots/all_projects_f1_boxplot.png	Summary F1 comparison across all projects